

Skills Summary

Place Value to 1,000: Three Ways to Write a Number

Use this reference while completing your worksheet or when studying.

Every number below is the same number three times. The place a digit sits in tells you what it is worth: hundreds, tens, ones. Once you know each digit's value, all three forms are just different ways of saying it.

1. Standard form to expanded form

Rule: give each digit its place value, then join them with +.

- hundreds digit $\times 100$ tens digit $\times 10$ ones digit $\times 1$
- A digit of 0 contributes nothing — leave that part out of the sum.

Example: Write 638 in expanded form.

Step 1. Expanded form asks for the value of each digit, so read the number one place at a time instead of as a whole.

Step 2. 6 is in the hundreds place $\rightarrow 600$; 3 is in the tens place $\rightarrow 30$; 8 is in the ones place $\rightarrow 8$.

$$638 = 600 + 30 + 8$$

Try it: Write 275 in expanded form.

$$275 = 200 + 70 + 5$$

2. Expanded form to standard form

Rule: each part of the sum fills exactly one place. Add them up, or just read off the leading digit of each part.

If a place is missing from the sum, that place gets a 0 — the zero holds the place open so the other digits do not slide over.

Example: Write $800 + 50 + 1$ in standard form.

Step 1. Each part names one place, so the job is to put one digit in each place rather than to do a long addition.

Step 2. 800 $\rightarrow$ 8 hundreds; 50 $\rightarrow$ 5 tens; 1 $\rightarrow$ 1 one.

$$851$$

Try it: Write $400 + 9$ in standard form.

$$409$$

3. Number names (word form) to standard form

Rule: break the name at the word “hundred”.

(*number*) *hundred* → the hundreds digit; the rest of the name is a two-digit number that fills the tens and ones places.

Teen and tens words are *one* two-digit chunk, not two digits written side by side: “fifteen” is 15, which is 1 ten and 5 ones.

Example: Write three hundred sixty-two in standard form.

Step 1. The name already tells you the place values, so split it at “hundred” and translate each chunk.

Step 2. “three hundred” → 300; “sixty” → 60; “two” → 2; so $300 + 60 + 2$.

362

Try it: Write nine hundred four in standard form.

check: 904

4. Comparing numbers written in different forms

Rule: put *both* numbers in standard form first, then compare place by place from the left.

Same hundreds? Compare tens. Same tens? Compare ones. Use $<$ for “is less than” and $>$ for “is greater than” — the small point aims at the smaller number.

Example: Which is greater: $500 + 20 + 7$ or five hundred seventy-two?

Step 1. The two numbers are written in different forms, so nothing can be compared until both are in standard form.

Step 2. $500 + 20 + 7 = 527$ and five hundred seventy-two = 572; both have 5 hundreds, and 2 tens is less than 7 tens.

$527 < 572$

Try it: Compare six hundred eight and $600 + 80$.

check: $608 > 680$

(!) Watch out: A zero in the middle is the trap. In 907 the tens place is empty, so the expanded form is $900 + 7$, not $90 + 7$. And in $600 + 5$ the missing tens part still needs a 0 in standard form: the answer is 605, never 65.